AYA EMAD ELSABY

Data Analyst | AI Engineer

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PROFESSIONAL SUMMARY

Data Analyst and AI Engineer with an engineering background in Navigation Science and Space Technology and a graduate of the DigiLians Data Analyst Diploma (MCIT & Military Academy). Builds end-to-end, data-driven solutions from raw data to actionable insights across space technology, healthcare, road safety, and business intelligence. Delivered predictive models with 98%+ accuracy, automated analytics workflows, and interactive decision-support dashboards. Microsoft Certified Power BI Data Analyst (PL-300); currently pursuing a Microsoft Data Engineering Diploma.

TECHNICAL SKILLS

Data Analysis & Engineering: SQL, Python (Pandas, NumPy), Advanced Excel (Pivot Tables, Complex Formulas), Data Cleaning, ETL, Statistical Analysis, Feature Engineering
Visualization & BI: Power BI (DAX, Power Query, Data Modeling), Tableau, Matplotlib, Seaborn, Automated Reporting
Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, PINNs, Predictive Modeling, Anomaly Detection, Explainable AI (SHAP)
Programming & Tools: Python (Advanced), R (Basic), Streamlit, FastAPI, GitHub
Soft Skills: Technical Leadership, Problem-Solving, Teamwork, Professional Communication
Languages: Arabic (Native), English (Professional)

PROFESSIONAL EXPERIENCE

AI Engineering Intern | AI Startup (Confidential) Jun 2025 – Present

- Collaborated with the development team to design and implement machine learning models built for startup scalability.
- Contributed to data preprocessing pipelines and model optimization to improve performance in real-world applications.

Junior Data Analyst | DigiLians, Ministry of Communications (MCIT) & Military Academy Jan 2025 – 2026

- Completed an intensive program covering the full data analysis lifecycle: data cleaning, statistical modeling, and predictive analysis; graduated with the Specialized Data Analyst Diploma.
- Applied Excel (Pivot Tables, complex formulas), SQL for relational databases, and Python (Pandas, NumPy, Matplotlib) to analytical projects.
- Built data visualizations in Tableau to communicate insights and support data-driven decision-making.
- Worked in a high-standard military-academic environment requiring discipline and professional communication.

Trainee | Micro Engineering Egypt (MEEG), METI Company Jul 2024

- Gained exposure to micro-engineering principles and technical project workflows in an industrial engineering context.
- Supported engineering teams with data-driven technical documentation and operational process optimization.

PROJECTS

AutoADHD – AI-Powered ADHD Screening & Clinical Decision Support GitHub
Built a clinical decision support system for ADHD screening using multimodal behavioral and neuroimaging data, covering preprocessing, feature engineering, dimensionality reduction, and predictive modeling. Developed an interactive application that stores patient records, provides AI-assisted assessments, and generates analytical reports for clinicians.
Tools: Python, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Streamlit, SHAP, PCA, FastAPI

Conjunction Risk Response & Satellite Maneuvers GitHub
Built a decision-support system that analyzes collision risks between satellites and orbital objects and optimizes maneuver strategies to ensure satellite safety.
Tools: Python, Orbital Mechanics Algorithms, Risk Assessment Models

ITC Egypt 2026 – Spectrum Anomaly Detection GitHub
Developed a machine learning solution to detect anomalies in radio-frequency spectrum data, flagging irregular signal patterns to support spectrum monitoring, for the ITC Egypt 2026 competition.
Tools: Python, Signal Processing, Machine Learning, Anomaly Detection

Predictive Maintenance System GitHub
Designed a machine learning model to monitor industrial equipment health and predict failures before they occur, reducing operational downtime.
Tools: Python, Scikit-learn, Pandas, Time-Series Analysis

Body Performance Analytics & Intelligent System GitHub

Analyzed body performance metrics and generated insights using intelligent data processing and machine learning techniques.

Tools: Python, Data Analysis, Machine Learning

Bank Management System GitHub

Developed a banking system simulation for managing accounts, transactions, and customer operations with a secure backend.

Tools: Python, OOP, File Handling / Database Concepts

EDUCATION

B.A. in Navigation Science and Space Technology | University of Beni Suef 2019 – 2024

- GPA: 3.77/4.0. First specialized college of its kind in the Middle East for astronautics and space applications.

Specialized Data Analyst Diploma (Graduated) | DigiLians, MCIT & Military Academy 2025 – 2026

- Professional certification in advanced data analytics and strategic reporting.

Microsoft Data Engineering Diploma In Progress

- Focus on scalable data pipelines and modern data engineering practices.

Diploma in Google Data Science | Digital Egypt Pioneers Initiative (DEPI) 2024 – 2025

- Data science methodologies and cloud-based analytics.

Data Science & Deep Learning Training | Zewail City of Science and Technology 2024

- Data science workflows and deep learning frameworks (TensorFlow, PyTorch).

AI & Machine Learning Training | Information Technology Institute (ITI) 2024

- Supervised and unsupervised learning, and NLP.

CERTIFICATIONS

- **Microsoft Certified: Power BI Data Analyst Associate (PL-300)** – Microsoft, 2026. Valid until July 2027. Credential ID: DF1453BEA35F2E4A
- **Google Data Analytics Professional Certificate** – Google & Coursera, April 2026 (data cleaning, SQL, Python, Tableau, statistical analysis, capstone project)
- **IBM Machine Learning & Deep Learning** – Machine Learning with Python; Introduction to Deep Learning & Neural Networks with Keras; Deep Learning with Keras and TensorFlow (2026)
- **Additional:** Introduction to Data Analysis Using Python (Google, 2026); Preparing Data for Analysis with Microsoft Excel (Microsoft, 2026); Delivering Quality Work with Agility (IBM, 2026)